Curriculum Vitae

Dave (Hyeonjung) Kim

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RESEARCH INTERESTS

Bioinspired Robots, Dexterous Manipulation, Tactile Sensing, Deep Reinforcement Learning for Control, Robot Learning

EDUCATION

University of California, Los Angeles *Los Angeles, CA, USA*

Doctor of Philosophy (Ph.D.) in Mechanical Engineering (Robotics, Systems & Control) *Fall 2026 – Present*

- Co-advised by Dr. Tyler Clites and Dr. Jonathan Hopkins
- Funded by Waldron Robotics Fellowship & Kwanjeong Educational Foundation Scholarship

Yonsei University *Seoul, Republic of Korea*

Bachelor of Science (B.S.) in Mechanical Engineering & Computer Science *March 2019 – February 2026*

- Cumulative GPA: 3.99/4.0, Summa Cum Laude
- 18-month absence for mandatory military service (vehicle mechanic)

RESEARCH EXPERIENCE

The Anatomical Engineering Group, *University of California, Los Angeles* *Los Angeles, CA, USA*

Graduate Student Researcher *September 2026 – Present*

- Developing methods to extend the loaded cycle life of novel robot finger joints
- Designing a tendon-driven robot hand for safe, compliant, contact-rich physical human-robot interaction
- Implementing hierarchical control across task, trajectory, and joint levels for dexterous manipulation

Biorobotics Lab, *Seoul National University (Advisor: Dr. Kyujin Cho)* *Seoul, Republic of Korea*

Undergraduate Researcher *August 2025 – July 2026*

- Designed a compact Dual-Transmission Actuator (D-TA) that enables adaptive switching between high-force and high-speed modes in tendon-driven systems
- Achieved nearly 10-fold increase in maximum grasp force while preserving grasp speed by integrating D-TA into tendon-driven prosthetic fingers
- Trained neural network mapping motor inputs and encoder readings to tendon tension for sensorless proprioceptive feedback

Human-Centered AI Robotics Lab, *Yonsei University (Advisor: Dr. Dongjun Shin)* *Seoul, Republic of Korea*

Undergraduate Researcher *December 2023 – July 2025*

- Developed a 3-DoF mass-spring-damper model of the Jump Enhancing Textile-Suit (JET-Suit) via system identification to predict assistive force as a function of actuation timing and participant characteristics
- Created an iterative decision process using the identified model to find each participant's optimal actuation timing for open-loop control
- Coordinated multi-session human subject study (N=12, 180+ jumps per participant), recording up to 5.52 cm vertical jump height increase while evaluating biomechanical adaptations

PUBLICATIONS & WORKING PAPERS

1. Sunghun Kim, Hyeokjun Seo, **Hyeonjung Kim**, Sengryeol Lee, Namho Kim, and Dongjun Shin, "A Soft Wearable Robot for Vertical Jump Enhancement via a Pneumatic Energy-Storing Propulsion Actuator and Triarticular Kinetic-Chained Structure," *Wiley Advanced Intelligent Systems* 8, e202500844 (2026). doi: 10.1002/aisy.202500844.

2. Sunghun Kim, Hyeokjun Seo, **Hyeonjung Kim**, Sengryeol Lee, Namho Kim, Dongjun Shin, "The JET Suit: A Soft Wearable Robot for Jump Augmentation and Neuromuscular Coordination Study," Accepted to Late-Breaking Results, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025*, Hangzhou, China.
3. Sunghyun Park, **Hyeonjung Kim**, "JOSE: Reconstructing Privileged State from Joint History for Humanoid Motion Control," Manuscript submitted to *IEEE International Conference on Robotics and Automation (ICRA) 2027*.
4. **Hyeonjung Kim**, Sunghyun Park, "JOSE: Must the IMU Be Sensed? Joint-History Reconstruction for Humanoid Control," Manuscript submitted to *Everything Beneath the Policy Workshop, CoRL 2026*.
5. **Hyeonjung Kim**, Seungmin Jeon, Jiwhan Bong, "SCOPE: Steer or Fine-tune? A Cheap Test for What a Frozen VLA Can Already Do," Manuscript in preparation.
6. **Hyeonjung Kim**, "A Multi-Power Source Dual-Transmission Actuator for Tendon-Driven Systems," Manuscript in preparation.

ENGINEERING PROJECTS

8th YAICON: RL for Zero-Demo VLA Fine-Tuning, *Yonsei Artificial Intelligence* *March 2026 – June 2026*

- Built a demonstration-free pipeline that trains a tool-latent-conditioned RL policy in Isaac Sim and auto-generates demos to fine-tune SmolVLA for general tool pick-place-handover
- Deployed the pipeline end-to-end, rolling out 500 episodes per tool to fine-tune SmolVLA for sim deployment on a SO-101 arm

7th YAICON: Sim2Real Transfer for Bipedal Locomotion, *Yonsei Artificial Intelligence* *June 2025 – November 2025*

- Developed custom LSTM-based risk-assessment reward for Adversarial Motion Prior (AMP)-based policy training for natural gait emergence on a 12-DoF bipedal humanoid
- Deployed learned locomotion policies on a physical Dynamixel-actuated platform

6th YAICON: YAKEMON Battle Frontier, *Yonsei Artificial Intelligence* *March 2025 – June 2025*

- Developed a custom Pokémon battle agent trained through self-play reinforcement learning with 96-dim state encoding
- Compared DQN, Dueling-DDQN, and PPO agents, with the Dueling-DDQN reaching a 91% peak win rate against the baseline while showing ability to exploit type matchups

Dr. CHA (Cleaning Hands-free, Any-wheel), *Creative Product Design (MEU4300)* *March 2025 – June 2025*

- Led development of autonomous wheelchair wheel cleaning robot with vision-guided positioning and app interface
- Contributed across design, modeling, control, Convolutional Neural Network implementation, and app development

Bibliobot, *Human Robotics (MEU5059)* *March 2025 – June 2025*

- Designed 5-DoF end-effector mechanism and derived forward and inverse kinematics for autonomous book retrieval
- Developed hierarchical control system spanning low-level joint control to safety-constrained trajectory planning

EXTRACURRICULAR ACTIVITIES

Yonsei Drone, *Mechanical Engineering Club* *January 2026 – July 2026*

- Participated in seminars on ROS and drone control
- Collected training data and implemented YOLO-based object detection on a fixed-wing UAV for the 24th Korea Robot Aircraft Competition

Yonsei Artificial Intelligence (YAI), *Academic Student Society* *December 2024 – July 2026*

- Served as executive staff, managing member attendance, seminar logistics, and club operations
- Led study groups on computer vision and reinforcement learning

RoboIn, *Mechanical Engineering Club* *June 2020 – January 2021*

- Participated in seminars on embedded systems and hardware design
- Designed posture feedback cushion using load cell array and developed SVM classifier for real-time sitting posture detection for the KU Medical Hackathon

Eagle Tutor, *Tutoring Program* *June 2020 – June 2025*

- Tutored 32 undergraduate students across *Engineering Mathematics I–IV* and *Mechanical Engineering Experiment I*

HONORS & AWARDS

Kwanjeong Educational Foundation Scholarship, *Kwanjeong Educational Foundation* *Fall 2026 – Fall 2028*
Funded a minimum of $25K annually for 5 years

Waldron Robotics Fellowship, *University of California, Los Angeles* *Fall 2026 – Fall 2028*
Full funding fellowship for two years

Summa Cum Laude, *Yonsei University* *February 2026*
Graduated in top 1% of cohort, College of Engineering

Creative Coursework Achievement Award, *Yonsei University College of Engineering* *Spring 2025*
For Dr. CHA developed for *Creative Product Design (MEU4300)*

University Placement (Jinri) Scholarship, *Yonsei University* *Fall 2019, Spring 2024*

Excellence Prize, KU Medical Hackathon, *Konkuk University Hospital* *November 2020*

Dean's List, *Yonsei University* *Spring 2019*

SKILLS & TECHNIQUES

Programming: Python, C++, C, Java, MATLAB/Simulink, LaTeX

AI & Robotics: PyTorch, TensorFlow, PyBullet, MuJoCo, Isaac Sim, Isaac Lab

Motion Capture & Biomechanics: Qualisys, Theia, Visual3D, OpenSim

CAD: SolidWorks, Creo, Autodesk Fusion

Embedded Systems: Arduino, ESP32, Raspberry Pi, Dynamixel

Languages: English (TOEFL 119, GRE 163V/170Q/5.0AW), Korean (Native)